

Plan

Tuesday, January 20, 2026 1:49 PM

CS-NETX 701 — Advanced Computer Networking (Network+ Mastery)

Textbook: Sybex *CompTIA Network+ Study Guide (N10-008)*

Total Chapters: 25

Course Length: 17 weeks (1 Spring Break)

Final Exam: Official **CompTIA Network+ (N10-008)**

Student: Single-student, mastery-paced

Pedagogical Structure

- Some weeks intentionally cover **2-3 tightly related chapters**
- Chapters are **grouped by conceptual cohesion**, not page count
- Labs + practice exams are derived directly from Sybex content
- Expansion material pushes well beyond certification level

Weekly Schedule (25 Chapters → 17 Weeks)

Week 1 — Foundations of Networking

■ Chapters:

- **Ch 1:** Introduction to Networks
- **Ch 2:** OSI Specifications

Core Concepts

- OSI vs TCP/IP (why both still matter)
- Encapsulation & abstraction failures
- Protocol layering as an engineering tool

Expansion

- OSI failure modes
- Cross-layer debugging logic

📝 Homework: Practice questions (Ch 1-2)

Week 2 — Physical Layer & Cabling

■ Chapters:

- **Ch 3:** Connectors & Wiring Standards
- **Ch 23:** Cable Connectivity Issues & Tools

Core Concepts

- Copper vs fiber physics
- EMI, attenuation, crosstalk
- Cable certification tools

Expansion

- Real-world cabling failures
- Enterprise wiring standards (TIA/EIA)

🔧 Lab: Cable fault diagnosis scenarios

📝 Practice questions

Week 3 — Ethernet Technologies

■ Chapters:

- **Ch 4:** Ethernet Specifications

Core Concepts

- Ethernet evolution
- Duplex, speed, auto-negotiation
- Frame structure

Expansion

- Performance tradeoffs
- Jumbo frames & MTU mismatches

🔧 Lab: Ethernet performance reasoning

📝 Practice questions

Week 4 — Networking Devices

■ Chapters:

- **Ch 5:** Networking Devices

Core Concepts

- Hubs vs switches vs routers
- Layered forwarding decisions
- ASIC vs software switching

Expansion

- Enterprise hardware design
- Failure domains

🔧 Lab: Device selection for scenarios

📝 Practice questions

Week 5 — IP Fundamentals

■ Chapters:

- **Ch 6:** Introduction to IP

Core Concepts

- IPv4/IPv6 structure
- Datagram forwarding
- Fragmentation

Expansion

- Path MTU discovery
- IPv6 transition mechanisms

🔧 Lab: Packet flow analysis

 Practice questions

Week 6 — IP Addressing

■ Chapters:

- Ch 7: IP Addressing

Core Concepts

- Address classes
- CIDR
- Private vs public addressing

Expansion

- Address planning for enterprises
- Address exhaustion strategies

 Lab: Addressing design

 Practice questions

Week 7 — Subnetting, NAT, IP Troubleshooting

■ Chapters:

- Ch 8: Subnetting, NAT, IP Troubleshooting

Core Concepts

- Binary subnetting
- NAT types
- IP failure diagnosis

Expansion

- NAT traversal issues
- Carrier-grade NAT

 Lab: Manual subnet derivation

 Practice questions

Week 8 — MIDTERM EXAM

■ Coverage:

- Chapters 1-8

Format

- Multiple-answer
- Scenario-based
- Short written reasoning

 45 minutes

 Weight: 20%

Week 9 — SPRING BREAK

- No new content
- Optional light review

Week 10 — Routing Fundamentals

■ Chapters:

- Ch 9: Introduction to IP Routing
- Ch 10: Routing Protocols

Core Concepts

- Static vs dynamic routing
- Distance vector vs link state
- Routing metrics

Expansion

- OSPF internals
- BGP conceptual overview

 Lab: Routing table analysis

 Practice questions

Week 11 — Switching & VLANs

■ Chapters:

- Ch 11: Switching & VLANs

Core Concepts

- MAC learning
- Broadcast domains
- VLAN trunking

Expansion

- VLAN hopping
- STP failure modes

 Lab: VLAN segmentation design

 Practice questions

Week 12 — Wireless Networking

■ Chapters:

- Ch 12: Wireless Networking

Core Concepts

- 802.11 standards
- RF behavior
- Wireless security

Expansion

- Site surveys
- Enterprise AP design

 Lab: Wireless troubleshooting

 Practice questions

Week 13 — Monitoring, Statistics & Sensors

 Chapters:

- **Ch 13:** Statistics & Sensors
- **Ch 22:** Ensuring Network Availability

Core Concepts

- SNMP
- Thresholds & alerts
- SLA concepts

Expansion

- Observability vs monitoring
- Telemetry pipelines

 Lab: Monitoring strategy design Practice questions**Week 14 — Policies, Availability & DR** Chapters:

- **Ch 14:** Organizational Documents
- **Ch 15:** High Availability & DR

Core Concepts

- Policies vs standards
- Redundancy models
- Backup strategies

Expansion

- RTO/RPO tradeoffs
- Business impact analysis

 Lab: DR planning exercise Practice questions**Week 15 — Security Fundamentals** Chapters:

- **Ch 16:** Security Concepts
- **Ch 17:** Common Attacks

Core Concepts

- CIA triad
- Threat vectors
- Attack taxonomy

Expansion

- Kill chain model
- Modern attack trends

 Lab: Threat modeling Practice questions**Week 16 — Network Hardening & Access** Chapters:

- **Ch 18:** Hardening Techniques
- **Ch 19:** Remote Access Security
- **Ch 20:** Physical Security

Core Concepts

- Defense-in-depth
- VPNs & zero trust
- Facility security

Expansion

- ZTNA vs VPN
- Insider threat mitigation

 Lab: Secure architecture design Practice questions**Week 17 — Data Centers, Tools & Final Prep** Chapters:

- **Ch 21:** Data Center & Cloud
- **Ch 24:** Troubleshooting Methodology
- **Ch 25:** Network Tools & Commands

Core Concepts

- Cloud networking models
- Structured troubleshooting
- CLI mastery

Expansion

- Hybrid cloud design
- Incident response workflows

 Full-length practice exams Target: 90%+**FINAL EXAM (Course Capstone)** **Official CompTIA Network+ (N10-008)**

- Pearson VUE (online or testing center)
- Serves as **course final**
- Passing = course completion

Outcome Guarantee (If Followed)

If the student completes:

- all practice exams,
- all labs,
- and remediation weeks,

then the **actual Network+ exam becomes procedural, not difficult.**

Next Steps (Optional Enhancements)

I can next:

- Build a **week-by-week lab manual**
- Create **exam-mapped Anki decks**
- Produce a **Pearson VUE scheduling plan**
- Generate a **midterm + mock final**

Just tell me which artifact you want next.

